

BRAC University
CSE320 DATA COMMUNICATIONS
Classwork: May 04, 2024, 23:59

Chapter(s): 6, 10, 12

Total Marks: 20

1. You have a "Hello" message to send to your enemy. To ensure your enemy knows if the message they received is correct, what will you really send them? You may use an ASCII table for this calculation. (Hint: Checksum Example 10.23) [5]
2. Discuss the advantages and disadvantages of DSSS. If the original signal is **1001**, and the spreading code is **0011001010**, draw the original signal, spread code and spread signal using unipolar NRZ and DSSS. (Hint: Figure 6.33) [5]
3. Name a MAC protocol that is used for
 - a. Guided Media
 - b. Ethernet
 - c. Radio Waves

How does the value of Back-off time (T_B) change if the number of attempts increases?

What are the possible Back-off times in the 5th attempt if the distance covered is 40 km and the signal speed is 13000 kms⁻¹? [5]

4. A common multipoint link is being used by four stations. Station 1 is to the furthest left and Station 4 is to the furthest right, with the remaining stations in between.
 - a. Station 1 attempts to send a frame to Station 4 at time $t = 10$ ms
 - b. The first bit of the frame reaches Stations 2, 3 and 4 at times $t = 11.7$ ms, 17 ms and 19 ms respectively
 - c. Station 1 sends the last bit of the frame at time $t = 27.3$ ms

Draw a space and time model of the channel if it is using CSMA/CD.

What is the Transmission Time and Maximum Propagation Time from the given information?

What should be the Minimum Frame Size and does it meet the requirements given above if the bandwidth is 7 Gbps? [5]